

Enhancing Accuracy of Operative Reports with Automated Artificial Intelligence Analysis of Surgical Video

Abhinav Khanna, MD, Tamir Wolf, MD, PhD, Igor Frank, MD, Alexander Krueger, Paras Shah, MD, Vidit Sharma, MD, Matthew T Gettman, MD, FACS, Stephen A Boorjian, MD, FACS, Dotan Asselmann, Matthew K Tollefson, MD

- BACKGROUND:** The creation of operative reports is a tedious documentation task that increases administrative burden, which is a potential driver of burnout. Additionally, operative reports are inherently subjective and may contain inaccuracies and incomplete information. Recent advances in artificial intelligence (AI) technology have enabled computer vision systems to accurately detect operative steps on surgical video. We aim to develop a platform for the automated creation of video-based AI surgical operative reports in robotic-assisted radical prostatectomy.
- STUDY DESIGN:** An AI computer vision algorithm was used to automatically detect surgical steps. Each step detected on video was mapped to prespecified text, which was then compiled into a narrative AI operative report. The accuracy of the AI operative reports was compared with operative reports written by surgeons, using expert review of raw surgical video footage as the ground truth.
- RESULTS:** A total of 158 cases from a tertiary referral center were included. In surgeon operative reports, 84 cases (53.2%) had at least 1 discrepancy between the operative report and surgical video. Of these, 43 cases (27.2%) had a clinically significant discrepancy based on expert video review. In AI operative reports, 46 cases (29.1%) had at least 1 discrepancy between the operative report and surgical video. Of these, 20 cases (12.7%) had a clinically significant discrepancy. Overall accuracy was higher for AI operative reports compared with surgeon operative reports (87.3% vs 72.8%, $p = 0.001$).
- CONCLUSIONS:** Operative reports created by AI achieved higher accuracy than those written by surgeons. To our knowledge, this is the first report of automated video-based AI surgical documentation. Future studies are warranted to explore the potential for this novel tool to reduce documentation burden, improve operative report accuracy, promote surgical transparency, and decrease subjectivity in surgical documentation. (J Am Coll Surg 2025;240:739–746. © 2025 by the American College of Surgeons. Published by Wolters Kluwer Health, Inc. All rights reserved.)

Operative reports are the only window into what transpires inside the operating room during surgery. They are the foundation for surgical documentation and are responsible for capturing the “what,” “why,” and “how”

of surgery. Operative reports not only facilitate communication between healthcare providers but also serve as the basis for surgical billing and coding, are used for surgical quality benchmarking, enable surgical research efforts,

Disclosure Information: Drs Wolf and Asselmann are employees of Theator, Inc.

Disclosures outside the scope of this work: Dr Sharma owns stock in and was a paid advisory board member at ImmunityBio and owns stock in MacroGenics. Dr Boorjian is a paid consultant to Ferring, ArTara, Prokarium, and Johnson and Johnson. Other authors have nothing to disclose.

Disclaimer: The commercial entity Theator Inc, assisted only with the application of artificial intelligence step-detection algorithm to the current data set. The commercial entity did not perform any of the data analysis, results interpretation, nor article preparation.

Received September 17, 2024; Revised November 4, 2024; Accepted November 5, 2024.

From the Department of Urology, Mayo Clinic, Rochester, MN (Khanna, Frank, Krueger, Shah, Sharma, Gettman, Boorjian, Tollefson); and Theator Inc, Palo Alto, CA (Wolf, Asselmann)

Correspondence address: Abhinav Khanna, MD, Department of Urology, Mayo Clinic, 200 1st St SW, Rochester, MN 55905. email: khanna.abhinav@mayo.edu

Abbreviations and Acronyms

AI	=	artificial intelligence
DVC	=	dorsal venous complex
RARP	=	robotic-assisted radical prostatectomy
VTN	=	video transformer network

and track compliance with regulatory requirements and evidence-based guidelines. Operative reports are arguably the single most important document in all of surgery.

However, recent literature has questioned the accuracy and utility of traditional surgical operative reports. Several studies have suggested that operative reports often contain inaccuracies and incomplete information and may not fully reflect what transpires during surgery.¹⁻⁵

Recent advances in computer vision technology have enabled the development of artificial intelligence (AI) algorithms for automatically detecting the operative steps being performed on raw surgical video footage. These recent advances lay the foundation to explore the use of AI in the creation of automated surgical operative reports. Herein, we aim to develop automated video-based AI operative reports in robotic-assisted radical prostatectomy (RARP), which is the most common robotic procedure performed in urology worldwide and is the mainstay of prostate cancer surgical treatment.⁶ Additionally, we aim to compare the accuracy of AI operative reports with those documented by surgeons in the medical record.

METHODS

Artificial intelligence operative report creation

We identified cases of RARP performed at an academic tertiary-care referral center from January 1, 2022, to December 31, 2022. Cases were included if full-length surgical video footage was available. Surgical video was shown to an AI computer vision algorithm previously developed by our group to perform automated detection of RARP surgical steps based on raw video footage.⁷ Briefly, the RARP step recognition AI algorithm employs 2 key elements. First, a visuospatial representation of each frame of surgical video was created using a deep feature extraction model. Second, surgical step prediction was performed on the basis of a learned sequence of features using a temporal model.⁷

For feature extraction, a video transformer network (VTN) was used, which analyzes the entire video as a series of image frames. VTN incorporates attention-based modules to capture both the spatial and temporal aspects of the video. The VTN model was fine-tuned using a novel RARP surgical video data set, leveraging knowledge from

other surgical types through transfer learning. After training, the VTN model functioned as a feature extractor for the RARP videos. The extracted features served as inputs for the next temporal model, which was a long short-term memory network.⁷

To ensure fidelity of the current data set, videos were excluded from this study if they had been previously used in the development of our previous RARP step-detection model. In this study, each surgical step detected by our AI algorithm was mapped to a prespecified string of text (Table 1). All text was joined together to create a narrative surgical summary, serving as an automated video-based AI operative report (Fig. 1). The electronic medical record was then reviewed to obtain operative reports written by surgeons, in which surgeons documented intraoperative findings and surgical techniques.

Assessing operative report accuracy

Surgeon operative reports were reviewed to identify the individual steps of RARP documented by surgeons in the medical record. AI operative reports were similarly assessed to identify the individual steps of RARP detected by our AI step-detection algorithm. All full-length surgical videos were independently reviewed by a team of human medical image annotators under the supervision of a fellowship-trained urologic oncologist to identify the surgical steps present (or absent) on surgical video footage. Human review of surgical video served as the ground truth for identifying individual steps performed during surgery. Predefined surgical steps included pelvic lymph node dissection, Space of Retzius dissection, dorsal vein ligation, anterior bladder neck transection, posterior bladder neck transection, seminal vesicle and posterior dissection, lateral/pedicle and apical dissection, urethral transection, vesicourethral anastomosis, and final inspection/extraction.

Surgeon and AI operative reports were each compared with the ground truth, which was derived from human review of surgical video footage. If the steps outlined in an operative report were in full agreement with those identified on surgical video, then the operative report was deemed accurate. If there was any discrepancy between the steps described in an operative report and those seen on video, then that surgical video underwent expert review by a urologic oncologist to adjudicate discrepancies. Discrepancies were categorized as either clinically significant or not clinically significant. Clinically significant discrepancies were defined as any of the following: (1) omission of key steps from the operative report that were seen on video (false negative), (2) inclusion of key steps in the operative report that

Table 1. Mapping of Individual Surgical Steps Performed During Robotic-Assisted Radical Prostatectomy to Narrative Text Describing Each Step in Video-Based Automated Artificial Intelligence Operative Report

Step	Text for artificial intelligence operative report
Preparation	Abdomen was insufflated and laparoscopic ports were placed
Lymph node dissection	Pelvic lymphadenectomy was performed, using the obturator nerve and external iliac vein as key landmarks
Retzius space dissection	The space of Retzius was developed
Anterior bladder neck transection	Anterior bladder neck was transected
Posterior bladder neck transection	Posterior bladder neck was transected
Seminal vesicles and posterior dissection	Vasa deferentia were divided and seminal vesicles were mobilized. A posterior plane was developed between the prostate and rectum from the prostate base to the apex
Lateral and apical prostate dissection	The lateral aspect of the prostate was then dissected. Prostatic vascular pedicles were ligated and neurovascular bundles were released. The apex of the prostate was then mobilized and care was taken to ensure the mobilization of the prostatic apex from the underlying rectum
Urethral transection	Urethra was divided
Vesicourethral anastomosis	Vesicourethral anastomosis was then performed
Final inspection and extraction	Surgical field was inspected. Specimen was placed into a laparoscopic specimen retrieval bag and extracted

were not seen on video (false positive), and (3) any error that implies a meaningful difference in technique (eg operative report describes anterior approach but video demonstrates Retzius-sparing technique, or operative report describes “hood sparing” but video demonstrates non-“hood-sparing” technique). Any discrepancy that did not meet these criteria was categorized as not clinically significant.

Overall accuracy, defined as the proportion of operative reports without a clinically significant discrepancy, was compared between surgeon and AI operative reports using a chi-square test. AI video analysis was performed using a cloud-based Health Insurance Portability and Accountability Act-compliant Surgical Intelligence platform (Theator Inc, Palo Alto, CA). This study was approved by the Mayo Clinic IRB.

RESULTS

Study cohort

A total of 158 RARP surgical videos from cases performed between January 2022 and December 2022 were included. The cohort included videos from 8 different surgeons at a single academic tertiary-care referral center.

Surgeon operative reports

A review of surgeon operative reports demonstrated that 84 cases (53.2%) had at least 1 discrepancy between the operative report and surgical video footage. Of these, 43 cases (27.2%) had a clinically significant discrepancy based on expert video review (Fig. 2). Clinically significant discrepancies included 37 cases in which the

surgeon operative report documented dorsal venous complex (DVC) ligation before transection, but video footage demonstrated DVC transection without previous ligation (“hood-sparing” technique). Conversely, 1 case was identified in which surgeon operative report documented DVC transection without previous ligation, but video footage demonstrated DVC ligation before transection. In 8 cases, pelvic lymphadenectomy was documented in the surgeon operative report, but a video review demonstrated that pelvic lymphadenectomy was not performed. Finally, 1 case was identified in which the surgeon operative report documented dissection of the Space of Retzius, but video footage demonstrated preservation of the Space of Retzius (“Retzius-sparing” technique). The median (interquartile range) word count for surgeon operative reports was 288.5 (279 to 347.8) words.

Artificial intelligence operative reports

Among AI operative reports, 46 cases (29.1%) had at least 1 discrepancy between the operative report and surgical video footage. Of these, 20 cases (12.7%) had a clinically significant discrepancy (Fig. 2). Clinically significant discrepancies were identified in several cases where the AI operative report failed to mention key surgical steps despite these steps being demonstrated on video footage. These omissions included 5 cases of anterior bladder neck transection, 5 cases of posterior bladder neck transection, 3 cases of final inspection, and 2 cases of pelvic lymphadenectomy that were seen on video but not included in AI operative reports. In 4 cases, the AI operative report

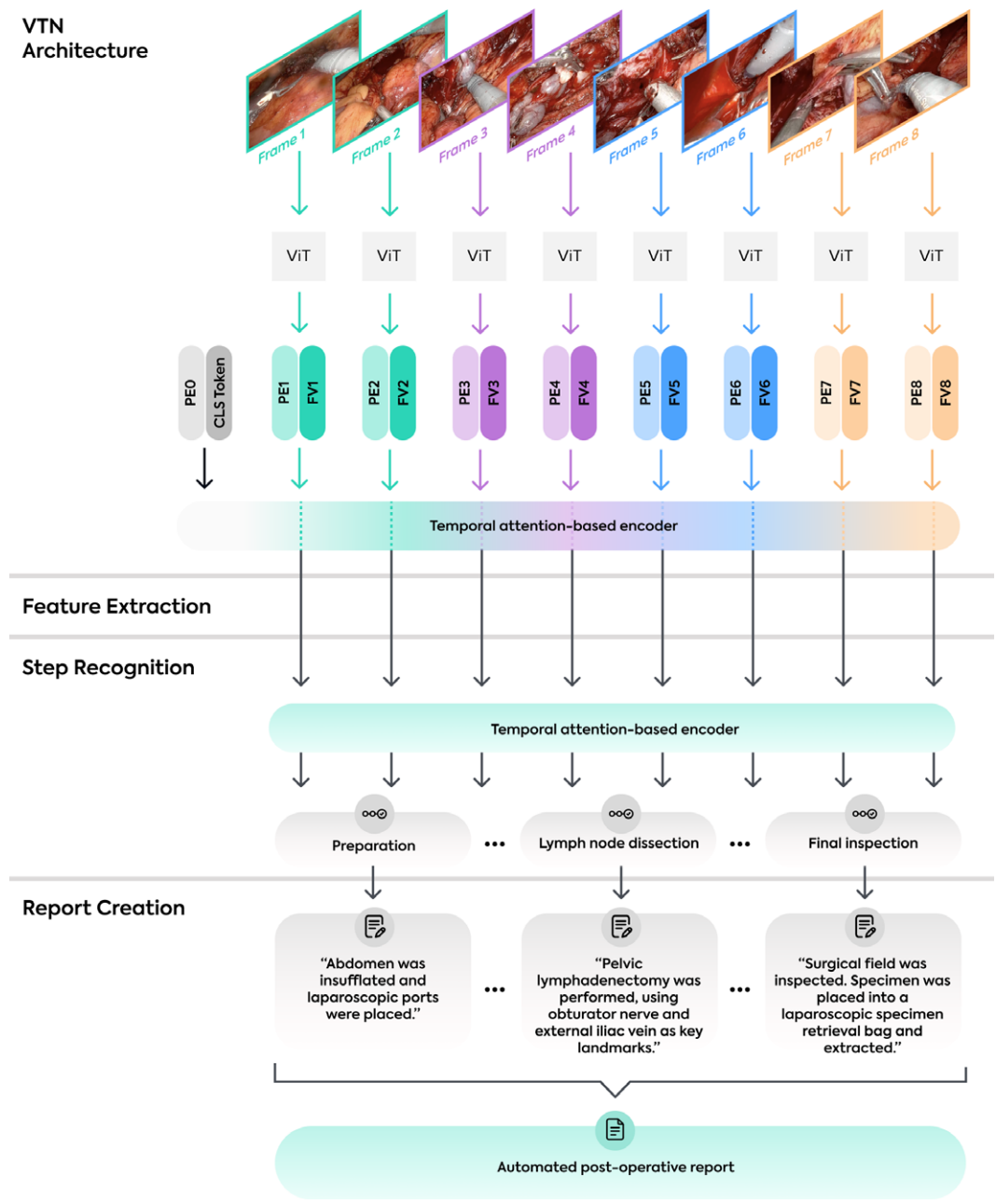


Figure 1. Machine learning pipeline for video-based automated artificial intelligence operative reports using only raw surgical video footage. CLS, classification; FV, feature vector; PE, positional embedding; ViT, vision transformer; VTN, video transformer network.

documented a pelvic lymphadenectomy, but an expert video review demonstrated that pelvic lymphadenectomy was not performed. Finally, in 7 cases, the AI operative report documented DVC transection before ligation (“hood-sparing” technique), but the video demonstrated DVC ligation before transection. The median (interquartile range) word count for AI operative reports was 132 (118 to 132) words.

Artificial intelligence vs surgeon accuracy

The total number of cases with at least 1 clinically significant discrepancy was 20 (12.7%) and 43 (27.2%) for the AI and surgeon operative reports, respectively. Overall accuracy was higher for AI operative reports compared with surgeon operative reports (87.3% vs 72.8%, $p = 0.001$). Table 2 highlights the specific clinically significant discrepancies observed in the surgeon and AI operative reports.

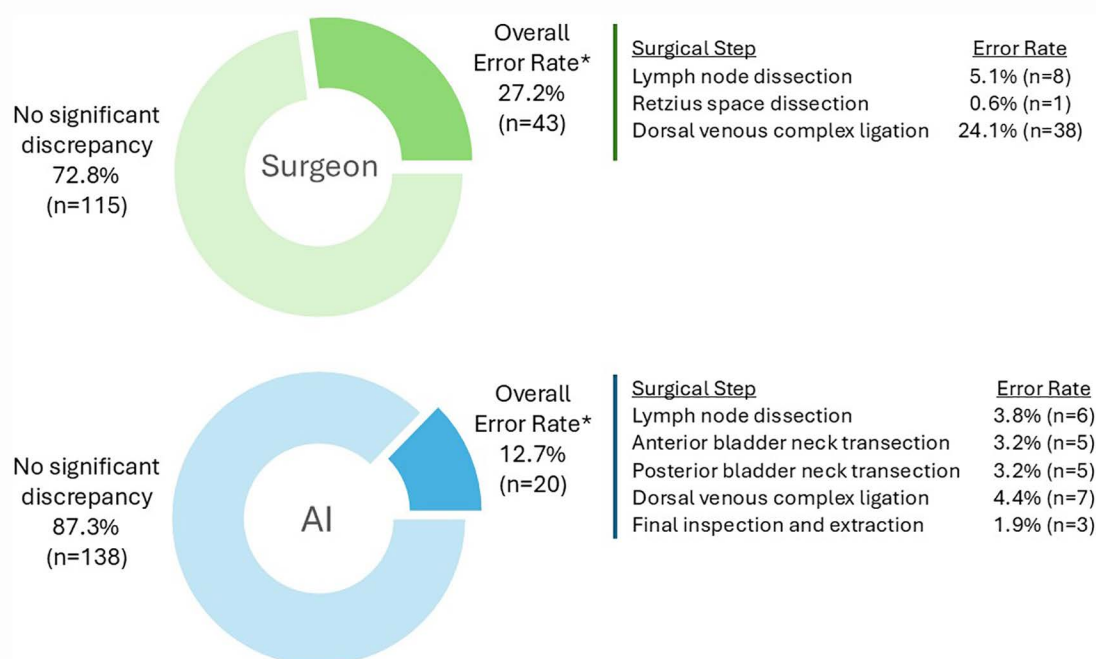


Figure 2. Accuracy of surgeon operative reports vs AI operative reports. *Includes any operative report with at least 1 clinically significant error. Some operative reports have more than 1 clinically significant error. Therefore, the overall error rate does not equal the sum of individual step error rates. p Value is 0.001 for the comparison of the overall error rate between surgeon vs AI operative reports. AI, artificial intelligence.

Table 2. Per-Step and Overall Clinically Significant Error Rate for AI Operative Reports vs Surgeon Operative Reports

Surgical step	AI error rate	Surgeon error rate
Lymph node dissection	6 (3.8)	8 (5.1)
Retzius space dissection	0 (0.0)	1 (0.6)
Anterior bladder neck transection	5 (3.2)	0 (0.0)
Posterior bladder neck transection	5 (3.2)	0 (0.0)
Dorsal venous complex ligation	7 (4.4)	38 (24.1)
Seminal vesicles and posterior dissection	0 (0.0)	0 (0.0)
Lateral and apical dissection	0 (0.0)	0 (0.0)
Urethral transection	0 (0.0)	0 (0.0)
Vesicourethral anastomosis	0 (0.0)	0 (0.0)
Final inspection and extraction	3 (1.9)	0 (0.0)
Overall error rate*†	20 (12.7)	43 (27.2)

Data presented as n (%).

*Any operative report with at least 1 clinically significant error. Some operative reports have more than 1 clinically significant error. Therefore, the overall error rate does not equal the sum of individual step error rates.

†p = 0.001 for comparison of overall error rate between AI and surgeon operative reports
AI, artificial intelligence.

DISCUSSION

We developed a platform for automated video-based AI operative reports in RARP. AI operative reports demonstrated higher overall accuracy than those documented by surgeons in the medical record. To our knowledge, this

represents the first-ever report of AI-automated surgical documentation in any surgical field.

The current paradigm of surgical documentation involves surgeons manually creating narrative summaries of surgical techniques and intraoperative findings.

This process is inherently subjective, laborious, and lacks transparency. Indeed, surgical operative reports have been demonstrated to be often incomplete or inaccurate across various surgical disciplines. In a cohort of patients undergoing laparoscopic cholecystectomy, Nijssen and colleagues¹ found that although 80.0% of surgeon operative reports documented the “critical view of safety” as being achieved, expert review of surgical video footage demonstrated that the critical view of safety was actually achieved in only 10.8% of cases. Additionally, among patients with intraoperative bile duct injuries, the critical view of safety was documented as achieved in 73.0% of surgeon operative reports, but a video review revealed it was actually achieved in zero cases.

Similarly, Wauben and colleagues² demonstrated that among all the surgical steps of laparoscopic cholecystectomy seen on surgical video, 26.2% were not described in the surgeon operative report. Notably, among cases in which intraoperative gallbladder perforation with spillage of bile and stones was seen on video, this adverse event was reported in only 73.3% of surgeon operative reports. Additional literature has described surgeon operative reports as being incomplete or inaccurate in pancreas surgery,³ head and neck surgery,⁴ and abdominal hernia surgery.⁵ Therefore, the inaccuracies observed in surgeon operative reports in this study are consistent with previous literature. Our results suggest that surgical video offers a potentially reliable and accurate alternative for documenting intraoperative surgical events, either independently or as a supplement to surgeon narratives.

In addition to being inherently subjective, the manual creation of operative reports is a tedious clerical task that contributes to the increasing administrative burden placed on physicians. Indeed, recent literature suggests medical documentation to be a primary driver of surgeon burnout.^{8,9} This study raises the possibility of automated surgical documentation in the future, the impact of which on surgeon burnout warrants further investigation.

Additionally, the surgeon operative report is currently the only window into what transpires inside an operating room and serves as the basis for surgical billing and coding. Automated operative reports may hold promise in shifting emphasis away from a surgeon's subjective recounting of events toward a more objective “ground truth,” such as surgical video footage. Indeed, in this study, there were 8 cases in which pelvic lymphadenectomy was documented in the surgeon operative report but not seen on surgical video footage. We surmise that this is likely related to the use of prepopulated templates within the electronic medical record, which have previously been suggested as a factor in medical documentation errors.^{10,11} Given that the inclusion of pelvic lymphadenectomy during RARP

has implications for billing and reimbursement,^{12,13} automated operative reports could lay the foundation for video-based surgical billing and coding in the future. AI operative reports warrant further study as a mechanism to improve billing accuracy, which further underscores the potential implications of this technology on surgical operations.

With regard to clinically significant discrepancies observed in surgeon operative reports, we cannot empirically assess the cause of these errors but hypothesize that these may be the result of prepopulated note templates. However, accurately capturing the relevant steps of surgery is clinically relevant and has downstream impacts on patient care. For instance, 8 surgeon operative reports in our cohort included inaccurate information about pelvic lymphadenectomy. In addition to the previously mentioned billing implications, pelvic lymphadenectomy provides important cancer staging and prognostic information and may potentially be therapeutic in certain subsets of men with high-risk prostate cancer.¹⁴ Therefore, there is clear clinical utility in accurately documenting whether lymphadenectomy was performed.

Another common area of surgeon operative report inaccuracy was related to descriptions of surgical technique. Recent literature has suggested that preservation of the anatomic support surrounding the urethra, known as the “hood-sparing” technique, may have implications for postoperative urinary incontinence, an important patient-reported outcome after RARP.¹⁵ This technique is characterized by intentional transection of the DVC before DVC ligation or, in some cases, avoidance of DVC ligation entirely, thereby preserving portions of the endopelvic fascia, detrusor apron, and puboprostatic ligaments.¹⁶ Therefore, this temporal relationship between DVC ligation and DVC transection during surgery can be used to signify the presence or absence of a hood-sparing surgical technique. Similarly, preservation of the Space of Retzius, known as the “Retzius-Sparing” technique, has also been implicated in promoting early return of urinary continence after RARP.¹⁷ We suspect that inaccuracies related to these nuances in surgical technique are unintended consequences of surgeons relying on note templates for surgical documentation.

However, the accurate description of surgical technique in the medical record is of paramount importance. Indeed, prospective randomized trials of surgical technique in any surgical field are sparse,^{18,19} so the overwhelming majority of literature to inform surgical techniques is retrospective and relies on chart review of operative reports. Unfortunately, our data would suggest that surgeon operative reports may be unreliable sources of truth about nuances in surgical technique. Therefore, there is significant potential for bias

in retrospective studies of surgical techniques, which are unable to ensure that the documented surgical technique being studied was actually performed during surgery. We believe automated operative reports based on surgical video footage warrant further exploration as the primary source of data in empiric evaluations of surgical technique.

In contrast to surgeon operative reports, we hypothesize that errors in AI operative reports were a result of inaccuracies in the underlying automated step-detection algorithm, which serves as the foundation for AI operative reports. Rather than systematic errors clustered around a specific step of surgery, the inaccuracies in AI operative reports were distributed across a variety of steps. However, the error rate of AI operative reports did not exceed 5% in any single step of RARP, and the overall accuracy of AI operative reports exceeded that of surgeon operative reports. These results highlight the promise of AI-generated operative reports as a potentially viable mechanism of automated surgical documentation in the future. Additionally, as AI computer vision technology advances rapidly, we predict that step-detection algorithm accuracy will continue to improve, thereby improving the utility of AI-generated surgical documentation.

For this article, we used robotic radical prostatectomy as a prototype because prostate cancer is the most common cancer among American men, and robotic prostatectomy is among the most common robotic cancer surgeries performed globally. However, this novel use of AI to generate surgical documentation based on surgical video footage is broadly applicable beyond radical prostatectomy. Indeed, our group has developed AI-automated step-detection algorithms across a multitude of surgery types in virtually every surgical subspecialty, and these algorithms are the fundamental engine that drive AI operative reports. Therefore, using this approach, AI operative reports are feasible in virtually any surgery that is performed using a “scope,” including most endoscopic, laparoscopic, and robotic procedures.

Although this study is the first report of using AI to automatically generate surgical documentation based on surgical video footage, similar technology has been applied in various other settings. AI has been explored for structuring and analysis of sports footage, security and surveillance video, education, and news.²⁰ Within medicine, several groups have explored the use of AI to automatically summarize audio from clinic-based encounters into outpatient clinical documentation.²¹⁻²³ Indeed, analogous to our current work in surgical documentation, efforts to automate medical office documentation using AI have potential to reduce clinician administrative burden while simultaneously increasing documentation accuracy and billing integrity. Although several commercially available products aim to automate outpatient clinic documentation using AI analysis of audio data (Abridge AI, Inc;

Augmedix, Inc; Nuance Communications, Inc), ours is the first exploration of applying this technology to video data in surgery, which remains a critical unmet need.

Study results should be interpreted in the context of methodological limitations. First, our current iteration of video-based AI operative report automation is focused on fundamental surgical steps of RARP but lacks the granularity to capture very subtle nuances in technique, such as the configuration of robotic ports or the specific types of sutures used. However, it is unclear whether such details warrant inclusion in surgical documentation. Second, due to the large dataset requirements for training AI step-detection models, our AI operative report is unable to capture and document rare variations in surgical anatomy, such as large prostatic median lobes or ectopic ureter insertion. However, surgical events that are agnostic to surgery type, such as hemorrhage events, can indeed be captured with our computer vision algorithm and incorporated into AI-generated operative reports. Third, our current AI operative report platform cannot articulate surgical complexity, such as inflamed or adherent tissue planes often seen with previously irradiated tissue or reoperative surgical fields. Finally, this study includes RARP cases from a single tertiary academic referral center and requires external validation.

Despite these limitations, this study represents the first-ever instance of video-based automated AI operative reports in surgery. This technology has the potential to serve as a catalyst for a future “digital” era of surgery, in which the ground truth of surgical documentation may not be based solely on subjective human narratives but may incorporate video footage of intraoperative steps, events, and key operative milestones.

CONCLUSIONS

In conclusion, we developed a platform for video-based automated AI operative reports in RARP. AI operative reports demonstrated higher accuracy than those documented by surgeons in the medical record. This novel application of computer vision technology marks the first foray into AI surgical documentation, which may have potential to improve documentation accuracy.

Author Contributions

Data curation: Khanna, Wolf, Frank, Krueger, Shah, Sharma, Gettman, Boorjian, Asselmann, Tollefson

Formal analysis: Khanna, Krueger, Sharma, Gettman, Boorjian, Asselmann, Tollefson

Investigation: Khanna, Wolf, Frank, Shah, Asselmann, Tollefson

Methodology: Khanna, Wolf, Frank, Krueger, Shah, Sharma, Gettman, Boorjian, Tollefson

Project administration: Khanna, Wolf, Shah, Sharma, Asselmann, Tollefson
 Resources: Khanna, Tollefson
 Supervision: Khanna, Asselmann, Tollefson
 Validation: Khanna, Boorjian
 Visualization: Khanna, Gettman, Tollefson
 Writing – original draft: Khanna
 Writing – review & editing: Khanna, Wolf, Frank, Krueger, Shah, Sharma, Gettman, Boorjian, Asselmann, Tollefson
 Software: Wolf, Asselmann
 Conceptualization: Khanna

Acknowledgment: We would like to acknowledge Dr Alenka Antolin, Dr Sari Maril, Dr Danielle Ben-Ayoun, and Maya Zohar for their contributions in data collection and article review.

REFERENCES

- Nijssen MAJ, Schreinemakers MJM, Meyer Z, et al. Complications after laparoscopic cholecystectomy: a video evaluation study of whether the critical view of safety was reached. *World J Surg* 2015;39:1798–1803.
- Wauben LSG, van Grevenstein WMU, Goossens RHM, et al. Operative notes do not reflect reality in laparoscopic cholecystectomy. *Br J Surg* 2011;98:1431–1436.
- Wiebe ME, Sandhu L, Takata JL, et al. Quality of narrative operative reports in pancreatic surgery. *Can J Surg* 2013;56:E121–E127.
- Nocon CC, Cohen MA, Langerman AJ. Quality of neck dissection operative reports. *Am J Otolaryngol* 2016;37:330–333.
- ten Broek RPG, van den Beukel BAW, van Goor H. Comparison of operative notes with real-time observation of adhesiolysis-related complications during surgery. *Br J Surg* 2013;100:426–432.
- Shah AA, Bandari J, Pelzman D, et al. Diffusion and adoption of the surgical robot in urology. *Transl Androl Urol* 2021;10:2151–2157.
- Khanna A, Antolin A, Bar O, et al. Automated identification of key steps in robotic-assisted radical prostatectomy using artificial intelligence. *J Urol* 2024;211:575–584.
- Senturk JC, Melnitchouk N. Surgeon burnout: defining, identifying, and addressing the new reality. *Clin Colon Rectal Surg* 2019;32:407–414.
- Rialon KL, Mueller C, Ottosen M, et al. Drivers of distress and well-being amongst pediatric surgeons. *J Pediatr Surg* 2021;56:841–848.
- Steinkamp J, Kantrowitz JJ, Airan-Javia S. Prevalence and sources of duplicate information in the electronic medical record. *JAMA Netw Open* 2022;5:e2233348.
- Bell SK, Delbanco T, Elmore JG, et al. Frequency and types of patient-reported errors in electronic health record ambulatory care notes. *JAMA Netw Open* 2020;3:e205867.
- Herrel LA, Syrjamaki JD, Linsell SM, et al. Identifying drivers of episode cost variation with radical prostatectomy. *Urology* 2016;97:105–110.
- Modi PK, Bock M, Kim S, et al. Utilization of pelvic lymph node dissection for patients with low-risk prostate cancer treated with robot-assisted radical prostatectomy. *Clin Genitourin Cancer* 2017;15:e1001–e1006.
- Lestingi JFP, Guglielmetti GB, Trinh Q-D, et al. Extended versus limited pelvic lymph node dissection during radical prostatectomy for intermediate- and high-risk prostate cancer: early oncological outcomes from a randomized phase 3 trial. *Eur Urol* 2021;79:595–604.
- Mian AH, Tollefson MK, Shah P, et al. Navigating now and next: recent advances and future horizons in robotic radical prostatectomy. *J Clin Med* 2024;13:359.
- Wagaskar VG, Mittal A, Sobotka S, et al. Hood technique for robotic radical prostatectomy-preserving periurethral anatomical structures in the space of Retzius and sparing the pouch of Douglas, enabling early return of continence without compromising surgical margin rates. *Eur Urol* 2021;80:213–221.
- Umari P, Eden C, Cahill D, et al. Retzius-sparing versus standard robot-assisted radical prostatectomy: a comparative prospective study of nearly 500 patients. *J Urol* 2021;205:780–790.
- Ramchurn TP, Nundy S. Randomised clinical trials in surgery: are we at a crossroads? *Ann Med Surg (Lond)* 2024;86:3–6.
- Robinson NB, Freme S, Hameed I, et al. Characteristics of randomized clinical trials in surgery from 2008 to 2020: a systematic review. *JAMA Netw Open* 2021;4:e2114494.
- Saini P, Kumar K, Kashid S, et al. Video summarization using deep learning techniques: a detailed analysis and investigation. *Artif Intell Rev* 2023;56:12347–12385.
- Kernberg A, Gold JA, Mohan V. Using ChatGPT-4 to create structured medical notes from audio recordings of physician-patient encounters: comparative study. *J Med Internet Res* 2024;26:e54419.
- Sain Buchem MM, Boosman H, Bauer MP, et al. The digital scribe in clinical practice: a scoping review and research agenda. *NPJ Digital Med* 2021;4:57.
- Wang J, Lavender M, Hoque E, et al. A patient-centered digital scribe for automatic medical documentation. *JAMIA Open* 2021;4:ooab003.

Invited Commentary

Beyond Matching: The Need for Context Awareness in Artificial Intelligence Analysis of Surgical Video

Daniel A Hashimoto, MD, FACS, Andrew J Hung, MD, Justin B Dimick, MD, MPH, FACS
 Ann Arbor, MI

There is a longstanding joke in computer science that artificial intelligence (AI) is the future—and always will be. Well, that joke is no longer funny. We are living in the era where AI is (and will continue) to transform how we live